

Basic Principles of Medicine 1
Module: Cardiovascular, Pulmonary,
Renal
Lecture: CPR 38

Gas Transport

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Learning Objectives

SOM.MK.II.BPM1.3.CPR.2.PHYS.0255	Define oxygen partial pressure (tension), oxygen content, and percent hemoglobin saturation as they pertain to blood.
SOM.MK.II.BPM1.3.CPR.2.PHYS.0256	Draw an oxyhemoglobin dissociation curve (hemoglobin oxygen equilibrium curve) showing the relationships between oxygen partial pressure, hemoglobin saturation, and blood oxygen content.
SOM.MK.II.BPM1.3.CPR.2.PHYS.0257	On the dissociation curve, illustrate the relationship between PO ₂ and dissolved plasma O ₂ content. Contrast the relative amounts of O ₂ carried bound to hemoglobin with that carried in dissolved form.
SOM.MK.II.BPM1.3.CPR.2.PHYS.0258	Describe how the shape of the oxyhemoglobin dissociation curve influences the uptake and delivery of oxygen and SaO ₂ .
SOM.MK.II.BPM1.3.CPR.2.PHYS.0259	Define P ₅₀ .
SOM.MK.II.BPM1.3.CPR.2.PHYS.0260	Show how the oxyhemoglobin dissociation curve is affected by changes in blood temperature, pH, PCO ₂ , and 2,3-DPG, and describe a situation where such changes have important physiological consequences.
SOM.MK.II.BPM1.3.CPR.2.PHYS.0261	Describe how anemia and carbon monoxide poisoning affect the shape of the oxyhemoglobin dissociation curve, PaO ₂ .
SOM.MK.II.BPM1.3.CPR.2.PHYS.0262	Name the factors that affect diffusive transport of a gas between alveolar gas and pulmonary capillary blood.

Learning Objectives

SOM.MK.II.BPM1.3.CPR.2.PHYS.0263	Describe the kinetics of oxygen transfer from alveolus to capillary and the concept of capillary reserve time (i.e., the portion of the erythrocyte transit time in which no further diffusion of oxygen occurs).
SOM.MK.II.BPM1.3.CPR.2.PHYS.0264	Define oxygen diffusing capacity and describe the rationale and technique for the use of carbon monoxide to determine diffusing capacity.
SOM.MK.II.BPM1.3.CPR.2.PHYS.0265	Calculate the diffusing capacity of carbon monoxide (DLCO) from the alveolar to arterial PO ₂ difference. Describe the normal value for DLCO and the significance of an elevated DO ₂ .
SOM.MK.II.BPM1.3.CPR.2.PHYS.0266	List the forms in which carbon dioxide is carried in the blood. Identify the percentage of total CO ₂ transported as each form.
SOM.MK.II.BPM1.3.CPR.2.PHYS.0267	Describe the importance of the chloride shift in the transport of CO ₂ by the blood.
SOM.MK.II.BPM1.3.CPR.2.PHYS.0268	Identify the enzyme that is essential to normal carbon dioxide transport by the blood and its location.
SOM.MK.II.BPM1.3.CPR.2.PHYS.0269	Explain the carbon dioxide dissociation curves for oxy- and deoxyhemoglobin. Describe the interplay between CO ₂ and O ₂ binding on hemoglobin that causes the Haldane effect.
SOM.MK.II.BPM1.3.CPR.2.PHYS.0270	Define respiratory acidosis and alkalosis and give clinical examples of each.

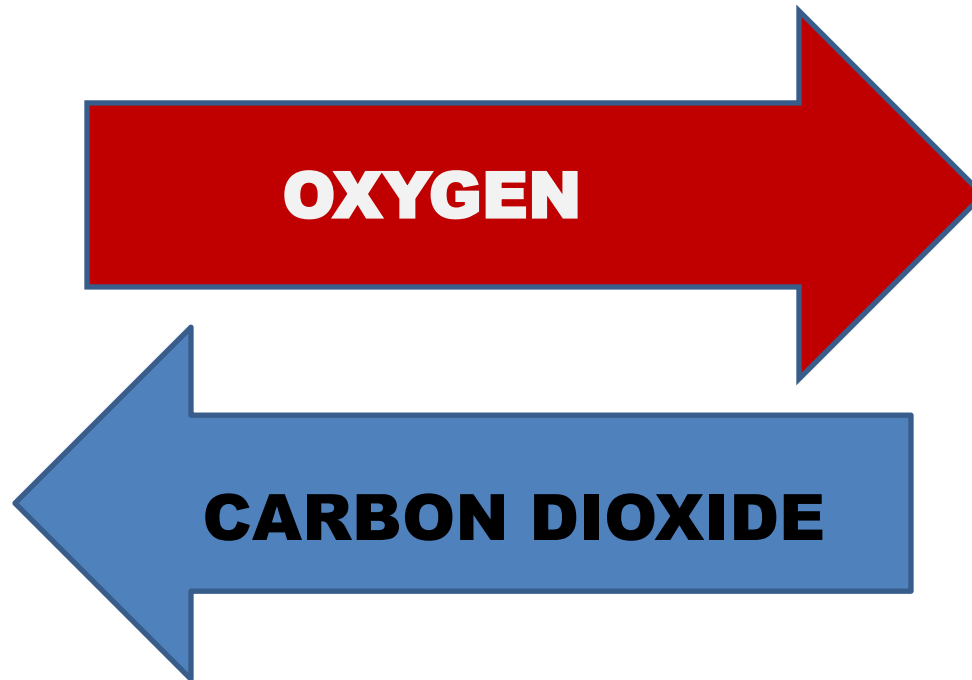
Recommended Reading

Rhoades and Bell, *Medical Physiology: Principles for Clinical Medicine*, Sixth Edition, © Lippincott, Williams & Wilkins (2023), Chapters 20

- <https://meded-lwwhealthlibrary-com.sgu.idm.oclc.org/content.aspx?sectionid=253088618&bookid=3211>

Slides with blue background are the supplemental information of the previous slides and thus, might not be explained again during the lecture time. However, please read the slides for clarification as they are equally important to know

Gas Transport

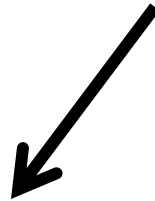


Gas exchange - the delivery of oxygen from the lungs to the pulmonary capillaries and the removal of carbon dioxide from tissue to the lungs. This movement is facilitated by **diffusion**.

Oxygen Transport

Oxygen Transport

Forms of Oxygen Transport



Hemoglobin

98% of O₂ transported in
blood (oxyhemoglobin)



Dissolved in Plasma

2% (0.3mL/dL)

Total blood O₂ content = Amount dissolved in plasma +
Amount bound to hemoglobin

Oxygen Transport

- Dissolved O_2 reaching circulation each minute is less than 10% of basal needs and would require a cardiac output of 83 L/min to maintain basal needs
- Dissolved O_2 is **not sufficient** to meet the body's demand.
- Blood O_2 content is increased by the carrier molecule – **hemoglobin**.

Oxygen Transport: Terminology

Important Terms

- **Oxygen Carrying Capacity**: the maximum amount of O_2 that **can be bound to** hemoglobin.
- **Oxygen Content**: amount of O_2 **actually bound to** hemoglobin + **dissolved** O_2
 - $O_2 \text{ content} = (\text{binding capacity} \times \% \text{ Saturation}) + \text{Dissolved } O_2$
- **Oxygen saturation**: **ratio** of quantity of O_2 actually bound to the quantity of O_2 that can be potentially bound ($O_2 \text{ content} / O_2 \text{ capacity}$)
 - Usually, 98% in arterial blood, on room air

Oxygen Transport

O₂ Carrying Capacity

- 1 g of Hb can bind 1.34 mL of O₂
 - Average Hb content of blood is 15 g/dL
 - 12-16 g/dL in females
 - 13-18 g/dL in men
- O₂ carrying capacity
= 1.34×15
= 20.1 ml O₂ /dL

O₂ Content

- O₂ content = (binding capacity × % Saturation) + Dissolved O₂

➤ O₂ content
= $(98\% \times 20.1) + 0.3$
= $19.7 + 0.3$
= 20.1 ml O₂ /dL

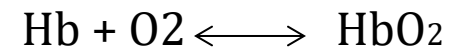
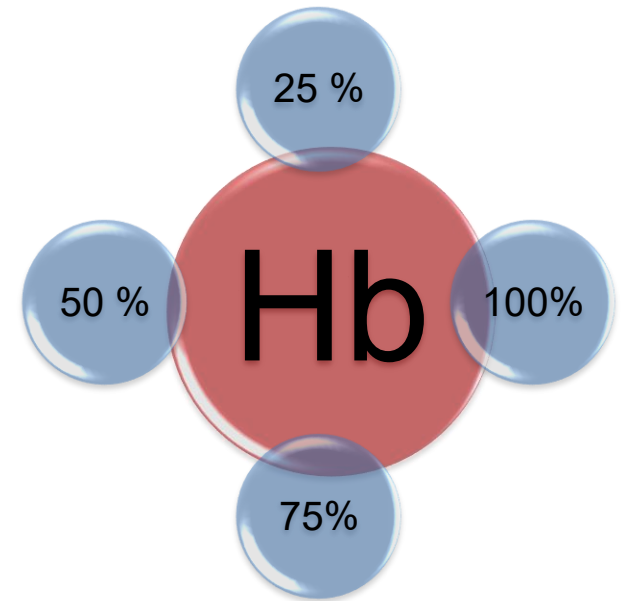
Oxygen Transport

O₂ Saturation (SaO₂)

- A measure of the number of occupied O₂ binding sites in a Hb molecule

$$\text{SaO}_2 = \frac{\text{HbO}_2 \text{ Content}}{\text{HbO}_2 \text{ Capacity}} \times 100$$

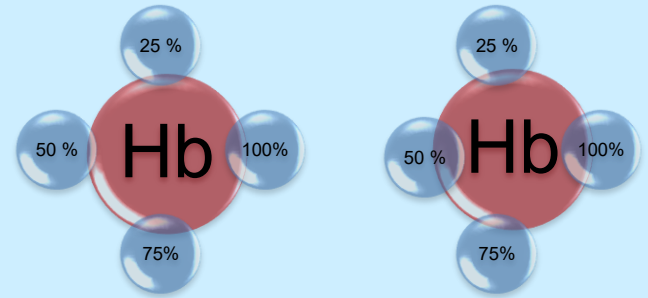
- A patient with anemia has a normal SaO₂ because the O₂ content and binding capacity are proportionally reduced



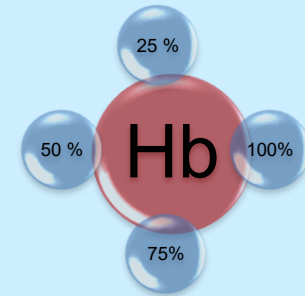
Oxyhemoglobin

Oxygen Transport: Anemia

- Anemia is a clinical condition of decreased red blood cell (RBC) mass
- May result from blood loss, increased destruction of RBCs (hemolysis), or decreased production of RBCs.
- The Hb available can still be saturated



2 saturated molecules
Content = 8 O₂



1 saturated molecule
Content = 4 O₂

Oxygen Transport

Hemoglobin-Oxygen Dissociation Curve

Association (loading)

- Blood arrives at lungs with a O_2 saturation of about 75%
- Pressure gradient enhances transfer
- Plateau region of the curve

Dissociation (unloading)

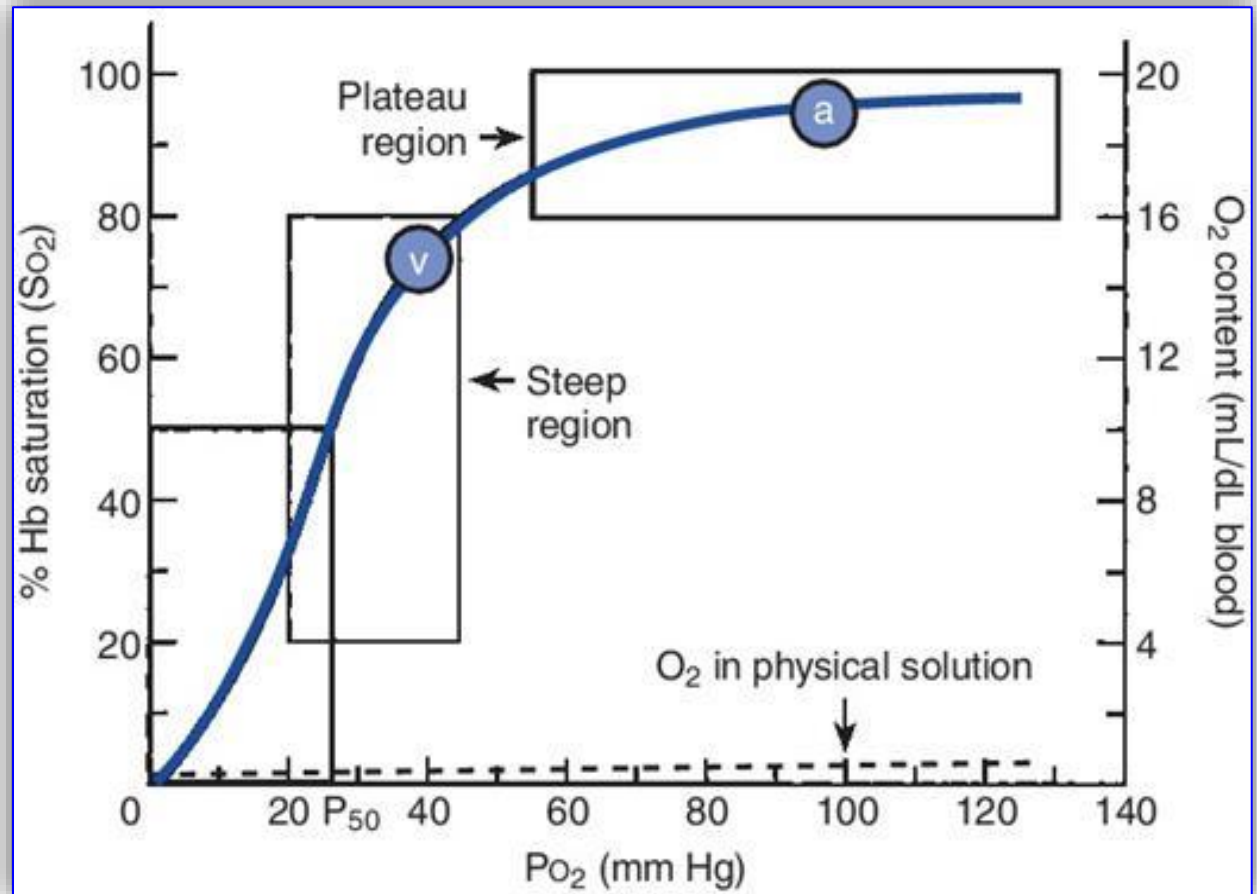
- Blood arrives at tissues where O_2 it is needed
- Pressure gradient and environment facilitates transfer
- Steep region of the curve
- Each O_2 dissociation event lowers the affinity of the remaining heme groups for bound O_2 .

Oxygen Transport

The Hemoglobin-Oxygen Dissociation Curve (oxyhemoglobin–equilibrium curve) shows the relationship between blood P_{O_2} , O_2 saturation, and oxygen content

a=arterial blood

v=venous blood



Oxygen Transport: P_{50}

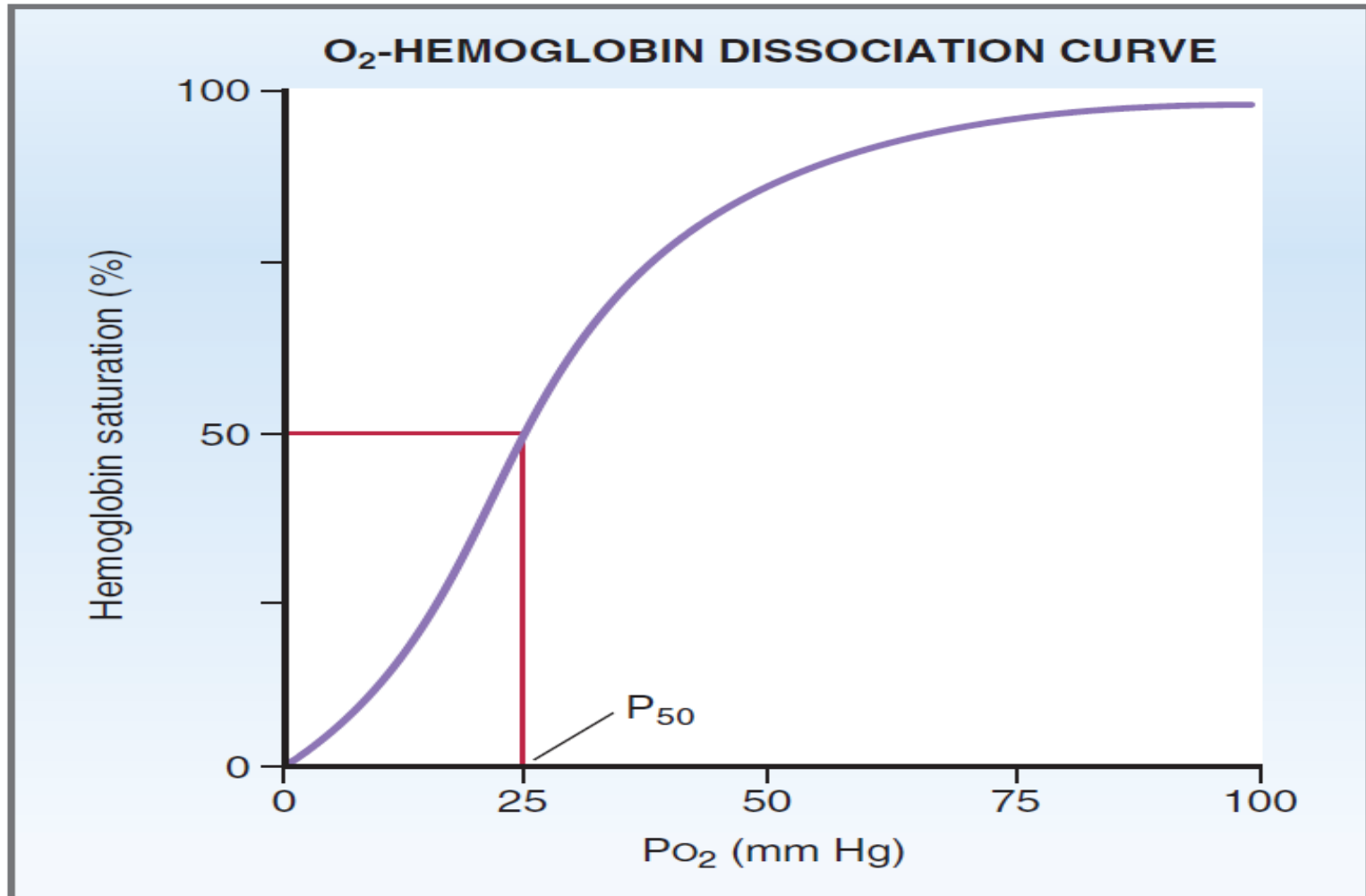


Figure 5-20 O₂-hemoglobin dissociation curve. P_{50} is the partial pressure of O₂ at which hemoglobin is 50% saturated.

Oxygen Transport: P_{50}

P_{50} : partial pressure of O_2 required to saturate 50% of Hb

- Measure of the affinity of Hb for O_2
- Low P_{50} = high affinity since less PO_2 is needed to reach 50% saturation
- High P_{50} = low affinity since you must give higher PO_2 to reach 50% saturation

Oxygen Transport

Hemoglobin-Oxygen Dissociation Curve

Rightward Shift (decreased O_2 affinity = offloading of O_2)

- Increased metabolism
 - Temperature increase
 - PCO_2 increase
 - H^+ increase
 - 2,3-bisphosphoglycerate increase (also a feature of chronic hypoxia - altitude, COPD, anemia)

Oxygen Transport

Hemoglobin- Oxygen Dissociation Curve

Leftward Shift (increased O_2 affinity = binding of O_2)

- Temperature decrease
- PCO_2 decrease
- H^+ decrease
- 2,3-Diphosphoglycerate decrease
- Carbon monoxide

Oxygen Transport

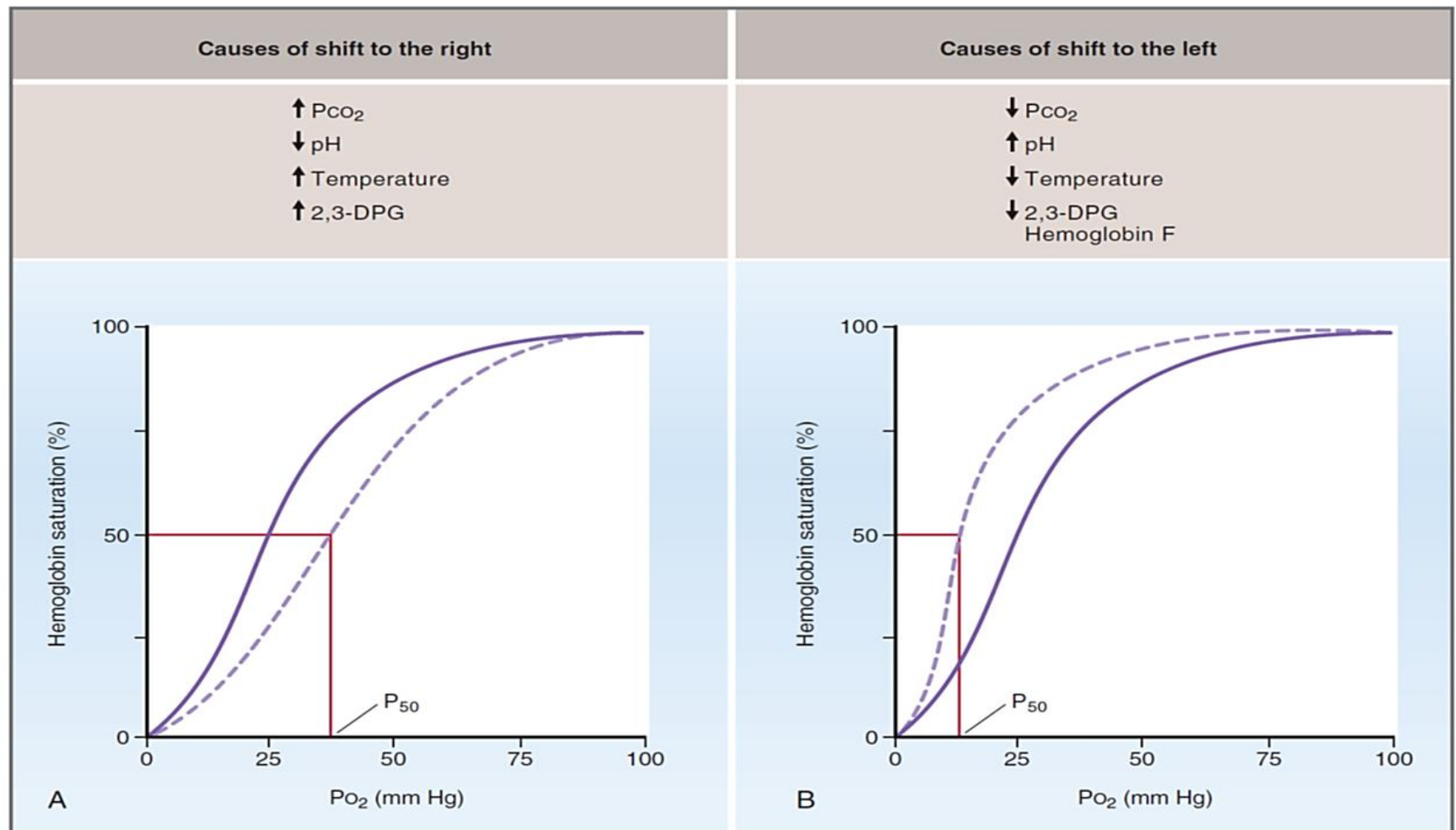


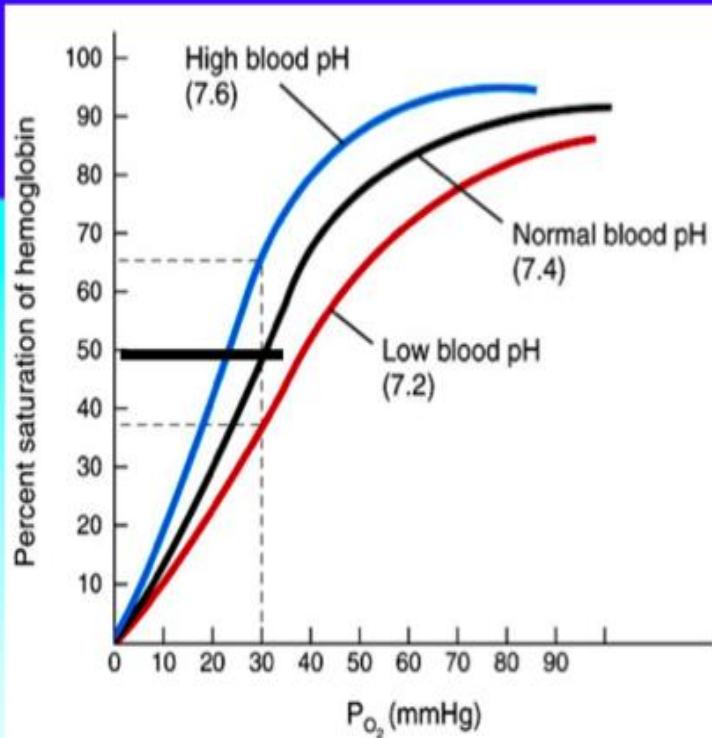
Figure 5-22 Shifts of the O_2 -hemoglobin dissociation curve. **A**, Shifts to the right are associated with increased P_{50} and decreased affinity. **B**, Shifts to the left are associated with decreased P_{50} and increased affinity.

Oxygen Transport

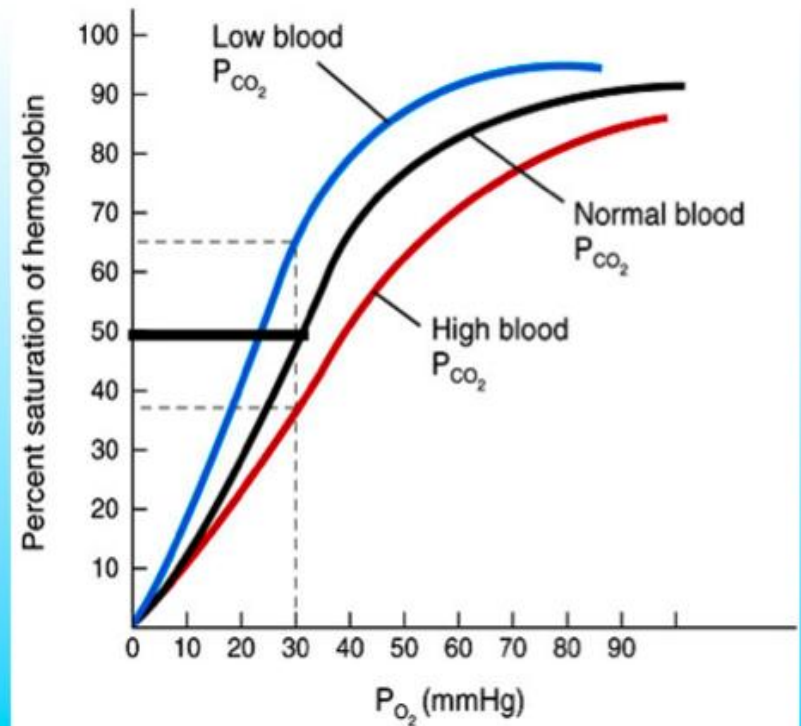
Bohr effect

- The effect of CO_2 and H^+ ions on Hb's affinity for O_2
- Elevation of those two parameters stimulate Hb to release oxygen.
- Right shift of the curve and an increased P_{50}

Oxygen Transport : Bohr effect



(a) Effect of pH on affinity of hemoglobin for oxygen



(b) Effect of P_{CO_2} on affinity of hemoglobin for oxygen

Bohr effect: Right shift of the curve with increased H^+ and PCO_2

Oxygen Transport in Carbon Monoxide (CO) Poisoning

- CO is colorless & odorless so undetectable
- PaO₂ is **normal** but O₂ content is **low**
- No increase in respiration to indicate a low O₂ content
- Skin may be cherry (bright) red but not cyanotic.

- Reduced O₂ content produces tissue hypoxia
- Symptoms include:
 - Headache, nausea,
 - Slow reaction time,
 - Blurred vision,
 - Confusion or coma.

Treatment:

- Administer 100% O₂ via nonrebreathing face mask
- Hyperbaric O₂ if severe

Oxygen Transport in Carbon Monoxide Poisoning

Carbon Monoxide

- Hb binds CO with high affinity (210 times greater than O₂)
- CO competes with O₂ for binding sites and increases the affinity of Hb for the bound O₂
- CO + Hb = Carboxyhemoglobin (bright red color)
- Deadly because O₂ is not released to the tissues

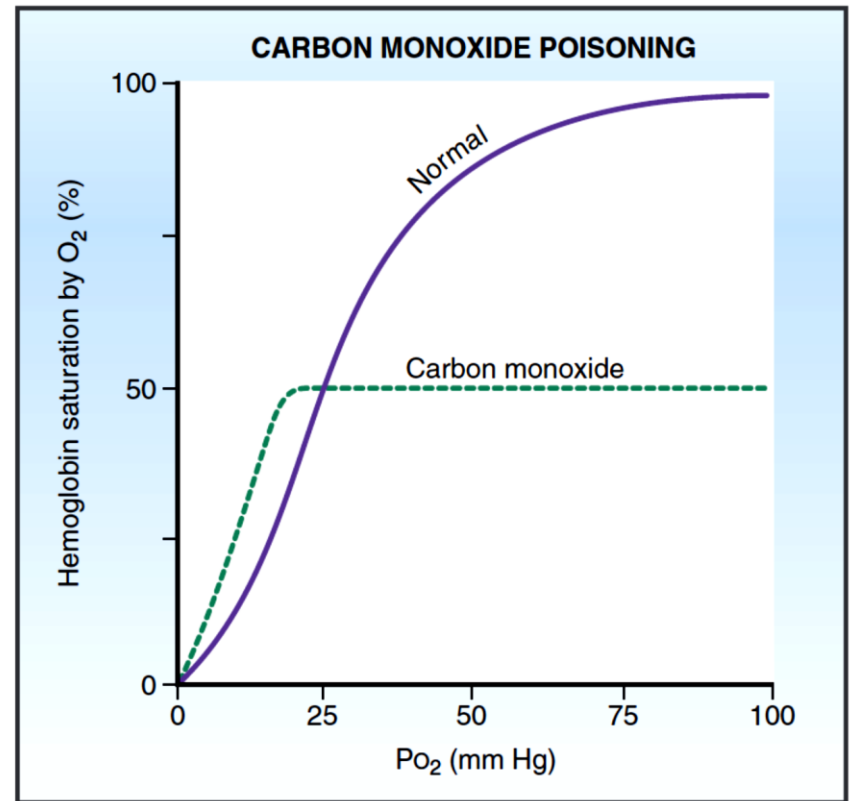
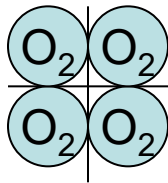


Figure 5-23 Effect of carbon monoxide on the O₂-hemoglobin dissociation curve. Carbon monoxide reduces the number of sites available for O₂ binding to hemoglobin and causes a shift of the O₂-hemoglobin dissociation curve to the left.

Oxygen Transport in Anemia

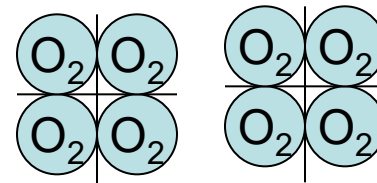
- Oxygen saturation of Hb **does not fall** because all O_2 binding sites are available and full (**SaO_2 normal**).
- **PaO_2** is also **normal** because P_aO_2 depends on dissolved O_2 and diffusion but not on O_2 bound to Hb.
- **Oxygen content falls** because Hb concentration of blood is low.

Patient 1 (Anemia)
has 1 Hb molecule



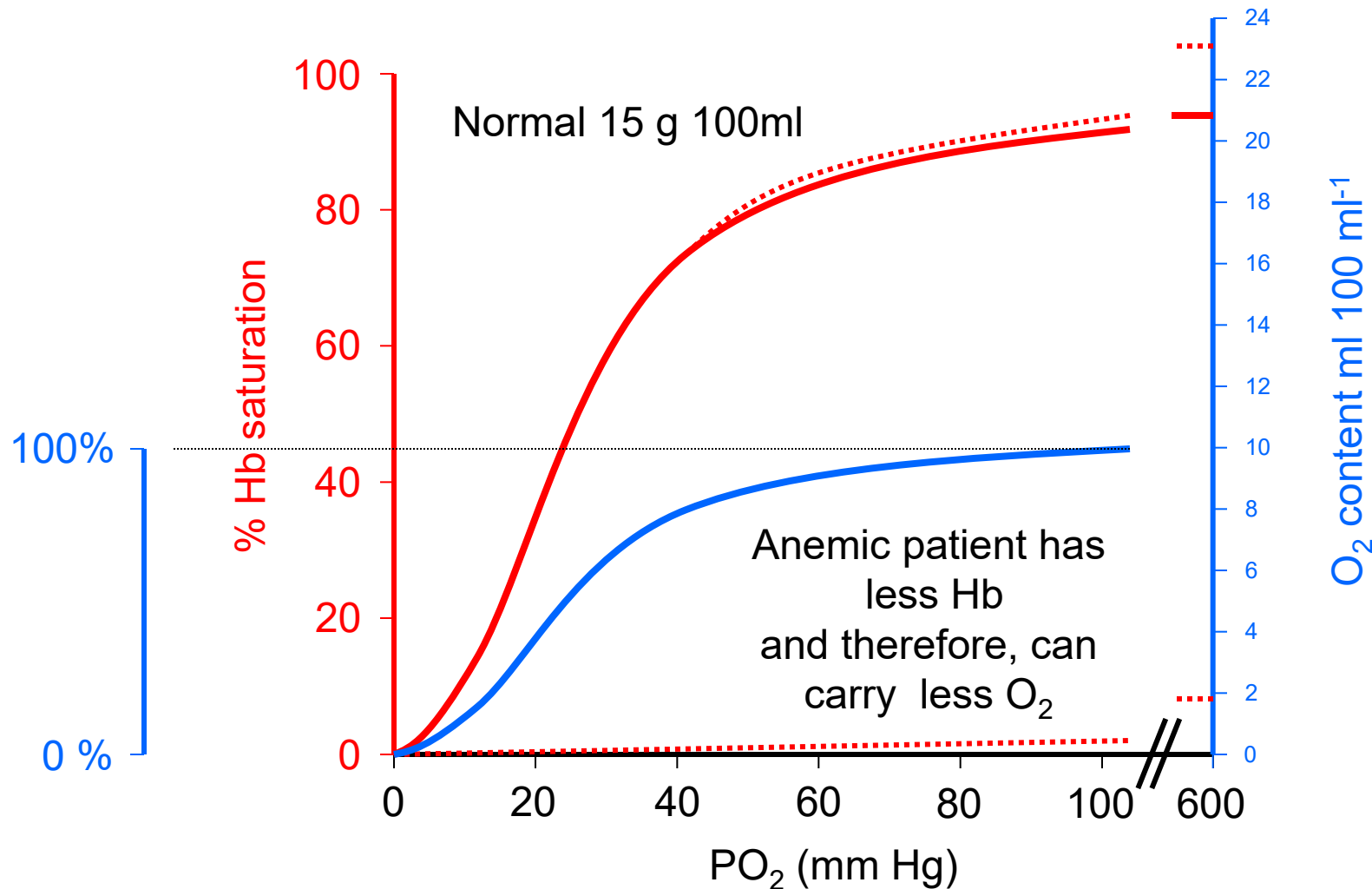
4 binding sites – 4 occupied
100% Sat
 O_2 content = 4 O_2

Patient 2 (Not anemic)
has 2 Hb molecules

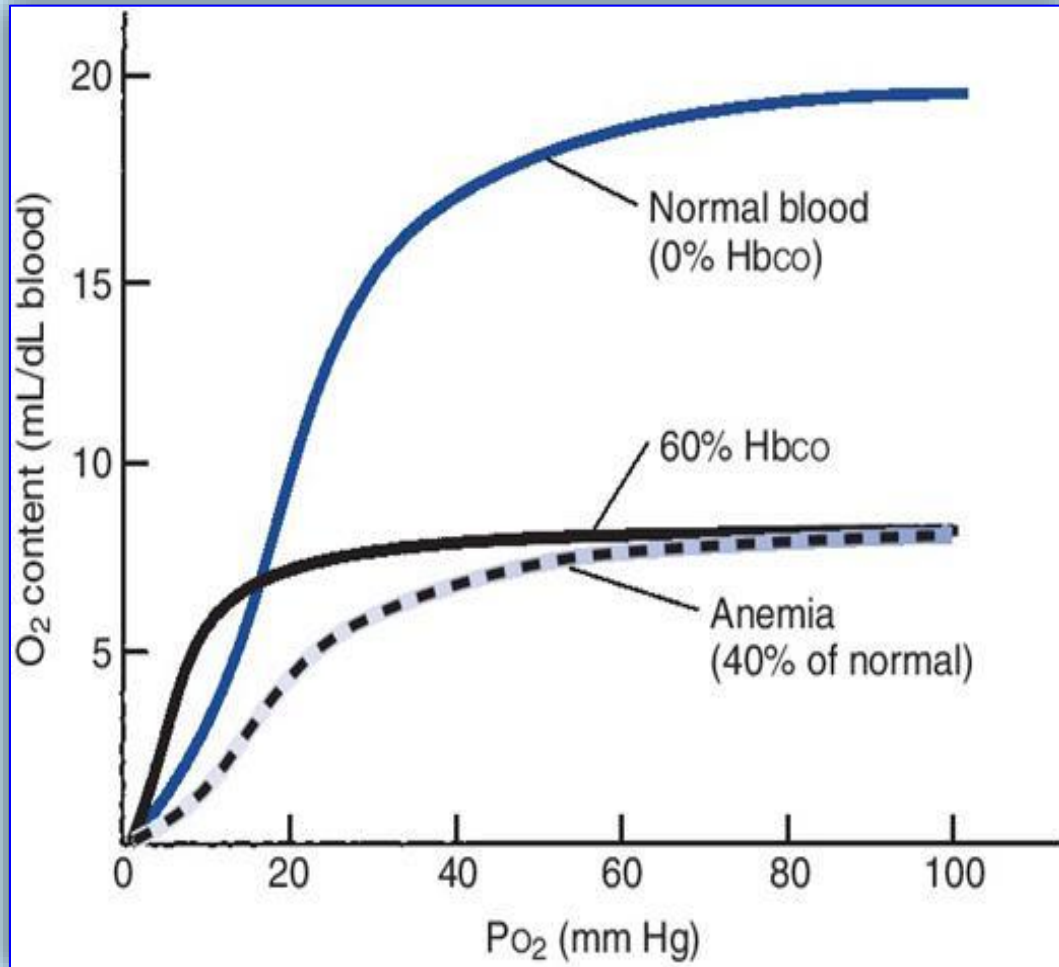


8 binding sites – 8 occupied
100% Sat
 O_2 content = 8 O_2

Oxygen Transport: Anemia O₂ saturation curve

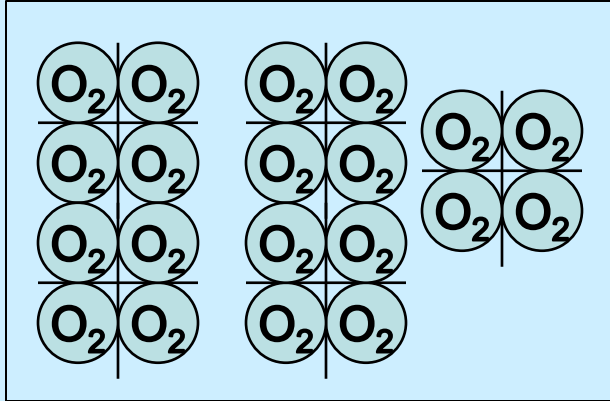


Hemoglobin-Oxygen Dissociation Curve in Anemia and CO poisoning



- A severe anemic patient has a lower the O_2 content (40% of normal).
- A patient whose Hb is 60% saturated with carbon monoxide (HbCO) also has a lower O_2 content. Additionally, the oxyhemoglobin curve is **shifted leftwards** when CO binds with hemoglobin.

Oxygen Transport in Polycythemia



- Polycythemia is a condition of an ↑ in red blood cells
- Therefore, Hb per 100 ml blood is also increased
- Increased oxygen carrying capacity of blood
- Increased oxygen content of blood
- No change in the O₂ affinity of Hb.

Polycythemia is associated with increased blood viscosity and vascular resistance that increases blood pressure.

Alveolar gas Exchange

Gas Transport & Exchange

There are 2 types of gas movements in the lungs

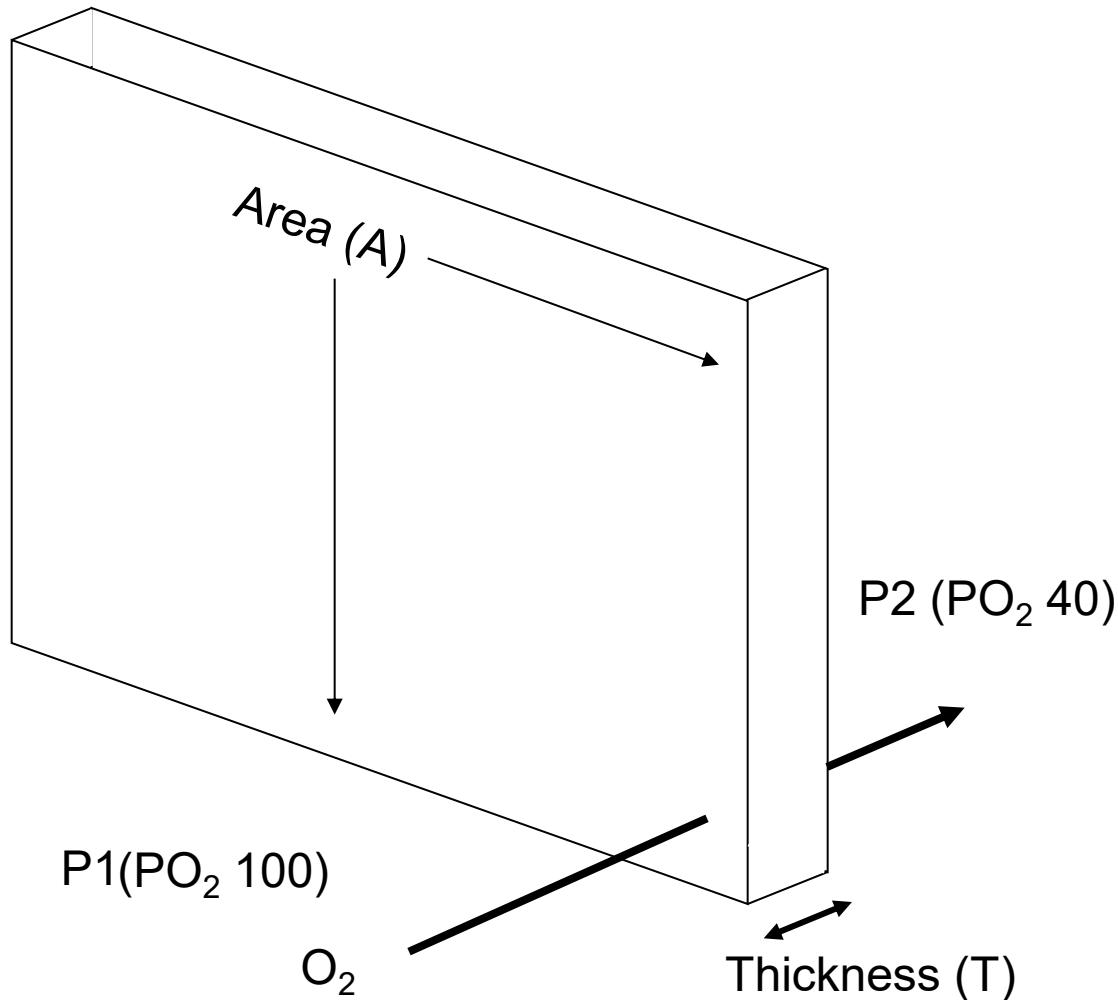
Bulk flow

- All gas molecules move as one unit
- That's how gas moves from **trachea to alveoli**
- Driving pressure is the pressure gradient between P_{atm} (at the mouth) – P_A .

Diffusion

- That's how gas moves from **alveoli to blood** or from **blood to tissues**
- Depends on:
Pressure gradient,
Membrane thickness,
Surface area, and
Diffusion coefficient.

Diffusion is governed by Ficks Law



Ficks Law

$$V_{\text{gas}} \propto A \cdot D \cdot (P_1 - P_2) / T$$

V_{gas} = volume of gas diffusing/min

Where:

A = surface area

D = solubility/ $\sqrt{\text{Mol wt}}$

P_1 & P_2 = partial pressure

T = thickness

$$V_{\text{gas}} = \frac{A \times D \times (P_1 - P_2)}{T}$$

$\frac{A \times D}{T}$ gives Diffusing capacity of lung (D_L) for that gas

Fick's law

- Rate of diffusion is **inversely related to the thickness** of the membrane
- If you double the thickness the rate of diffusion would be halved
- It is **directly proportional to surface area**
- If you double the surface area the rate of diffusion will double

Diffusion coefficient (D) =
solubility/ $\sqrt{\text{Mol wt}}$

- more soluble the gas
& < the molecular weight
> the diffusion

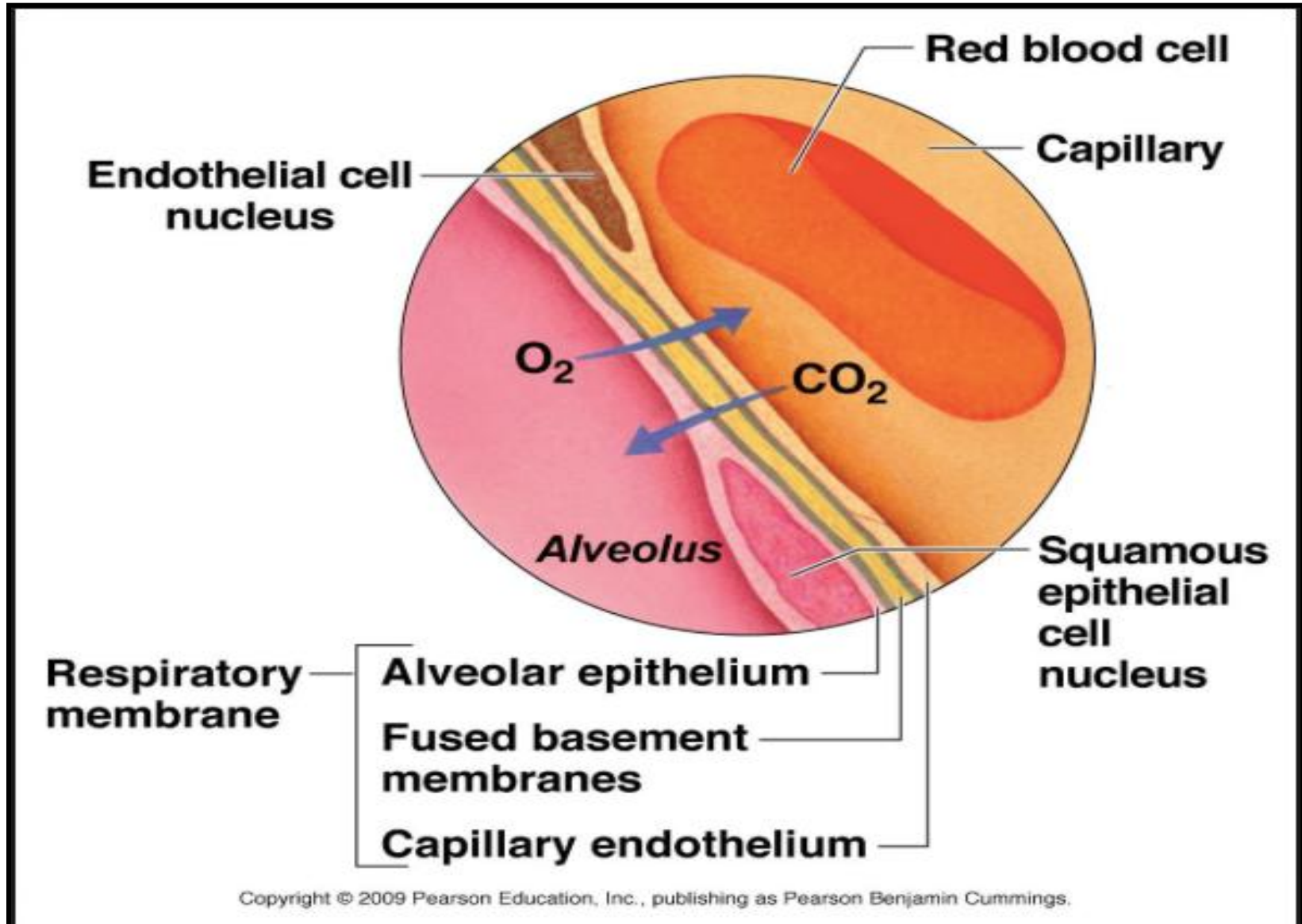
Rate of O₂ diffusion =
250mL/min

Rate of CO₂ diffusion =
200mL/min

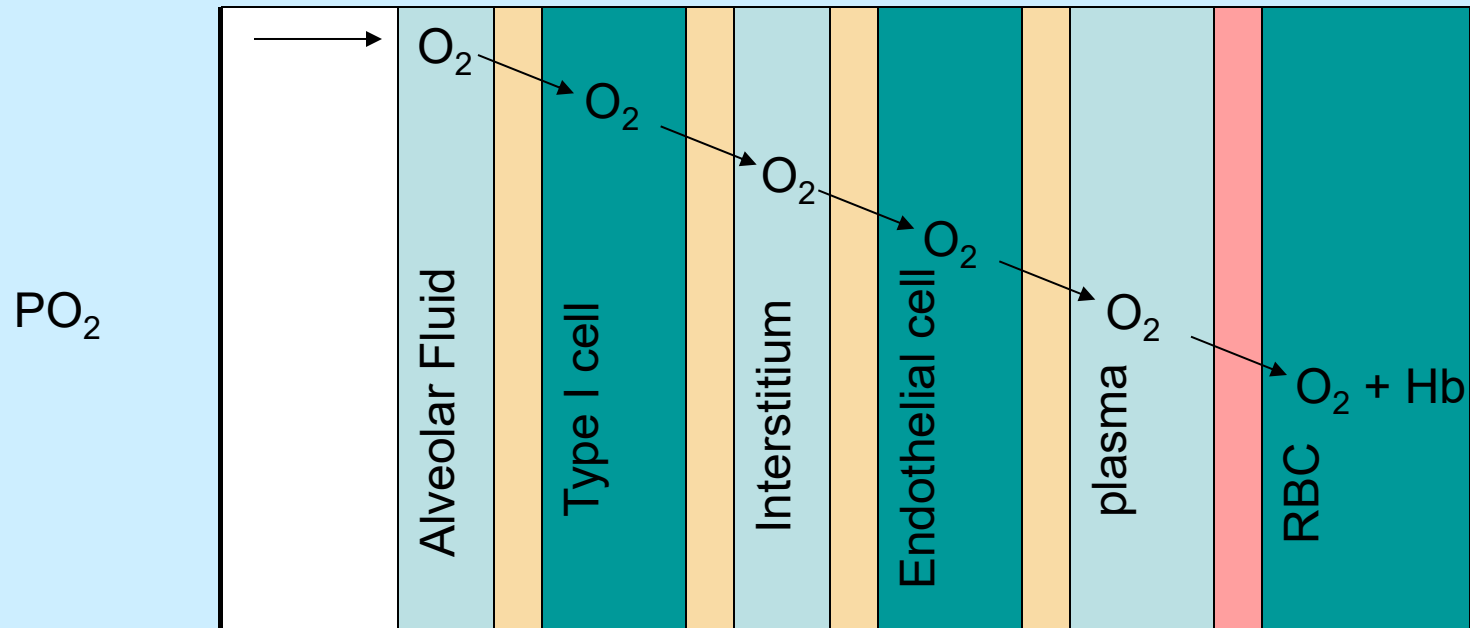
$R = V\text{CO}_2 / V\text{O}_2 = 0.8$

R- respiratory exchange ratio

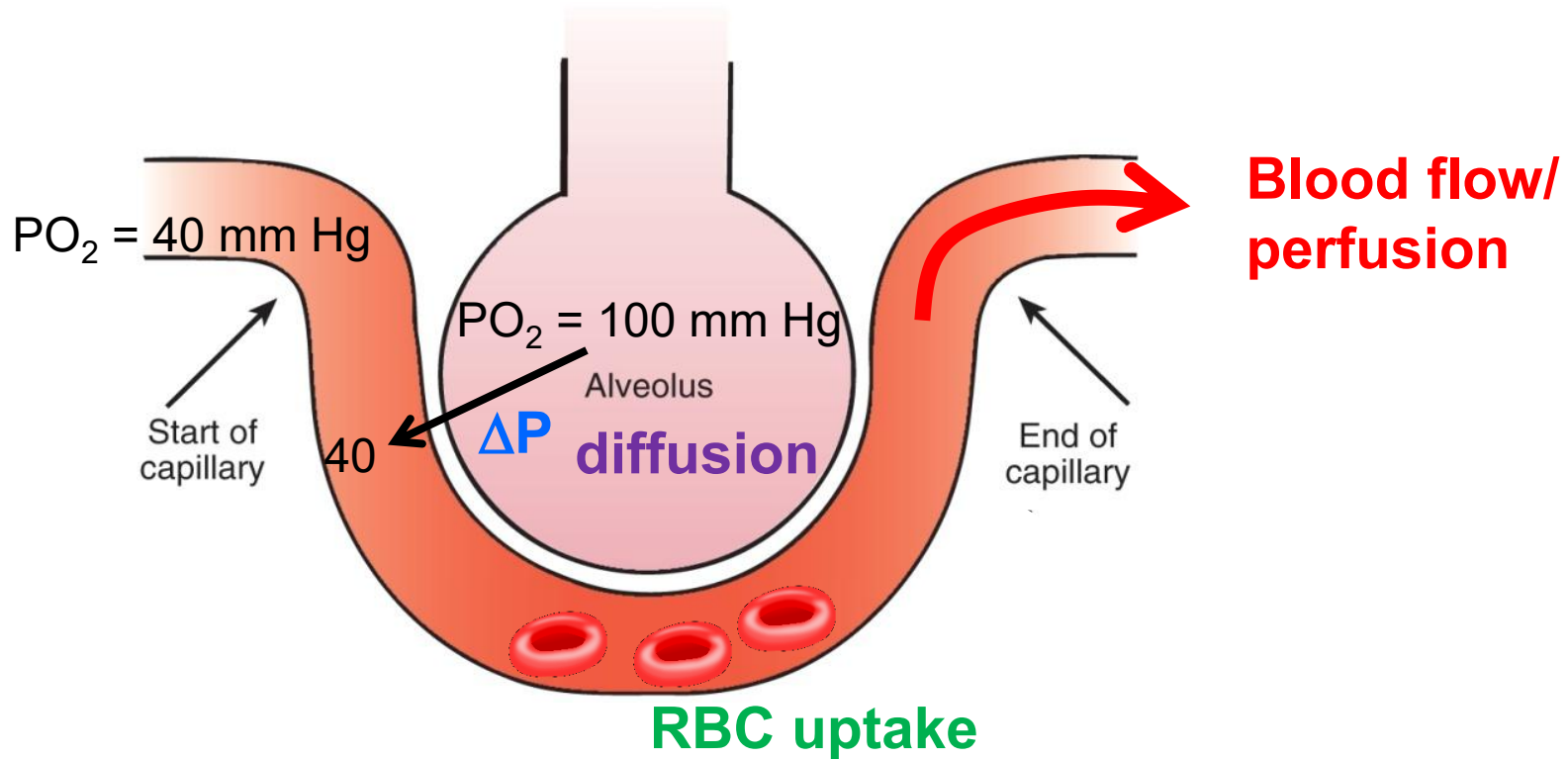
Diffusion Barrier



Diffusion Barrier

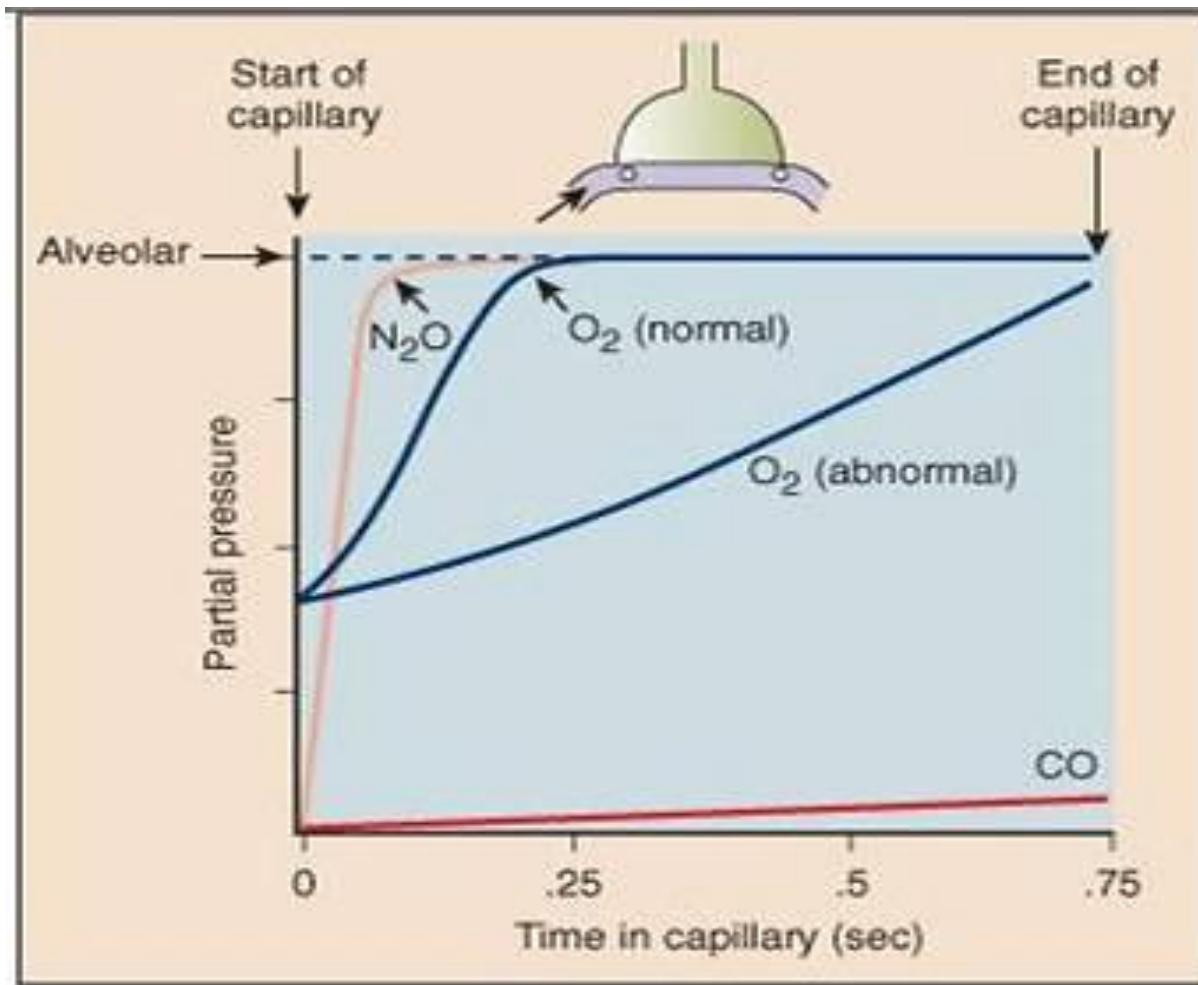


Getting oxygen from alveoli to blood



- Time required for RBC to move in the alveolar capillary is 0.75 secs at rest.
- This is the time during which the blood must equilibrate with alveolar gas.
- It is known as **transit time** which can vary with cardiac output (during exercise transit time is decreased due rapid blood flow).
- Amount of O_2 uptake is limited by pulmonary capillary blood flow

Effect of Blood Flow on Uptake Different Gases



Equilibration times

- N₂O - 0.10 secs
- O₂ - 0.25 secs
- CO does not equilibrate

Transit time never drops below 0.25 s therefore blood PO₂ is usually not affected (healthy individuals)

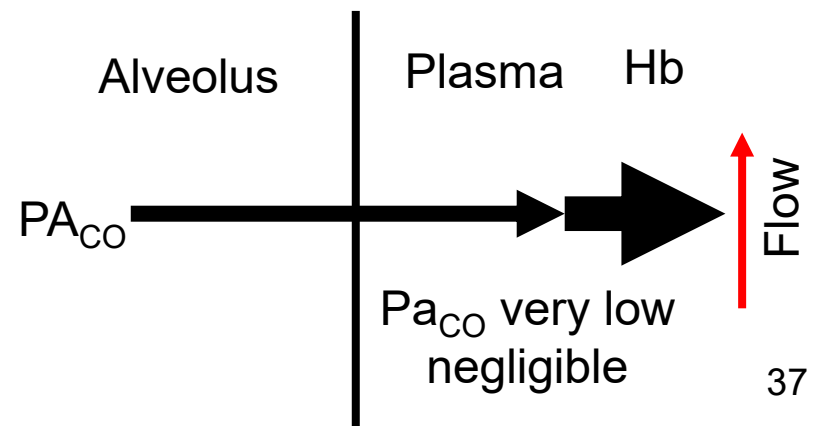
N₂O: Perfusion-limited vs CO: Diffusion-limited

- Is a dental anesthetic
- Does not bind to Hb
- It rapidly equilibrates (in 0.1 sec) with pulm. cap blood.
- Once it equilibrates, its partial pressure gradient is zero. That means no more N₂O can diffuse into blood.
- Its diffusion is increased by an increase in blood flow.

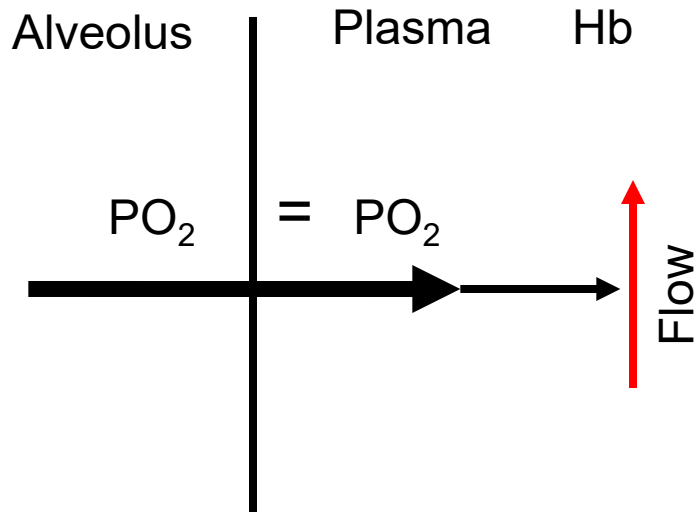
This is **perfusion-limited** gas

- CO binds rapidly to Hb (210 times faster than O₂)
- However, its rate of diffusion through the capillary barrier is slower
- It takes >0.75 secs to equilibrate with pulm capillary blood.
- So, there is constant partial pressure gradient between PACO & PaCO (pulm capillaries)

This is **diffusion-limited** gas



Oxygen: Perfusion-limited

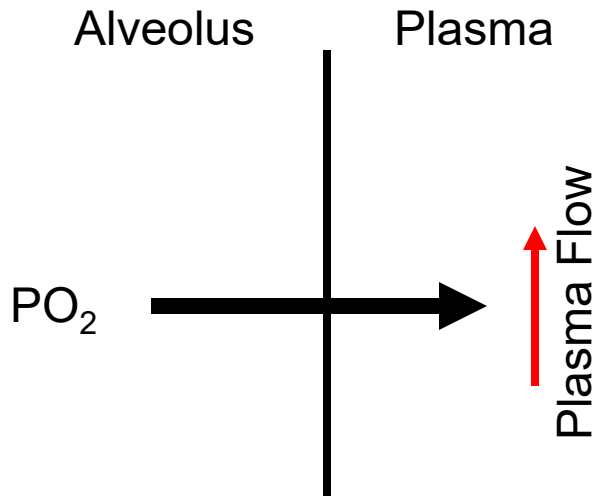


- O_2 diffuses rapidly across the barrier has affinity for Hb but $< CO$
- The rate of this reaction is slower than CO, however its diffusion across the barrier allows it to equilibrate quickly in 0.25 secs
- It has 0.5 secs reserve time

Oxygen is therefore perfusion-limited, normally.

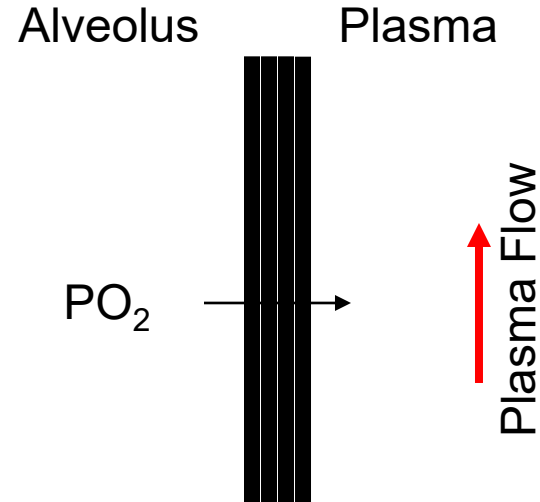
O₂ Perfusion/Diffusion-Limited

Normal



O₂ is Perfusion-limited in a **healthy normal person** because PO₂ equilibrates rapidly

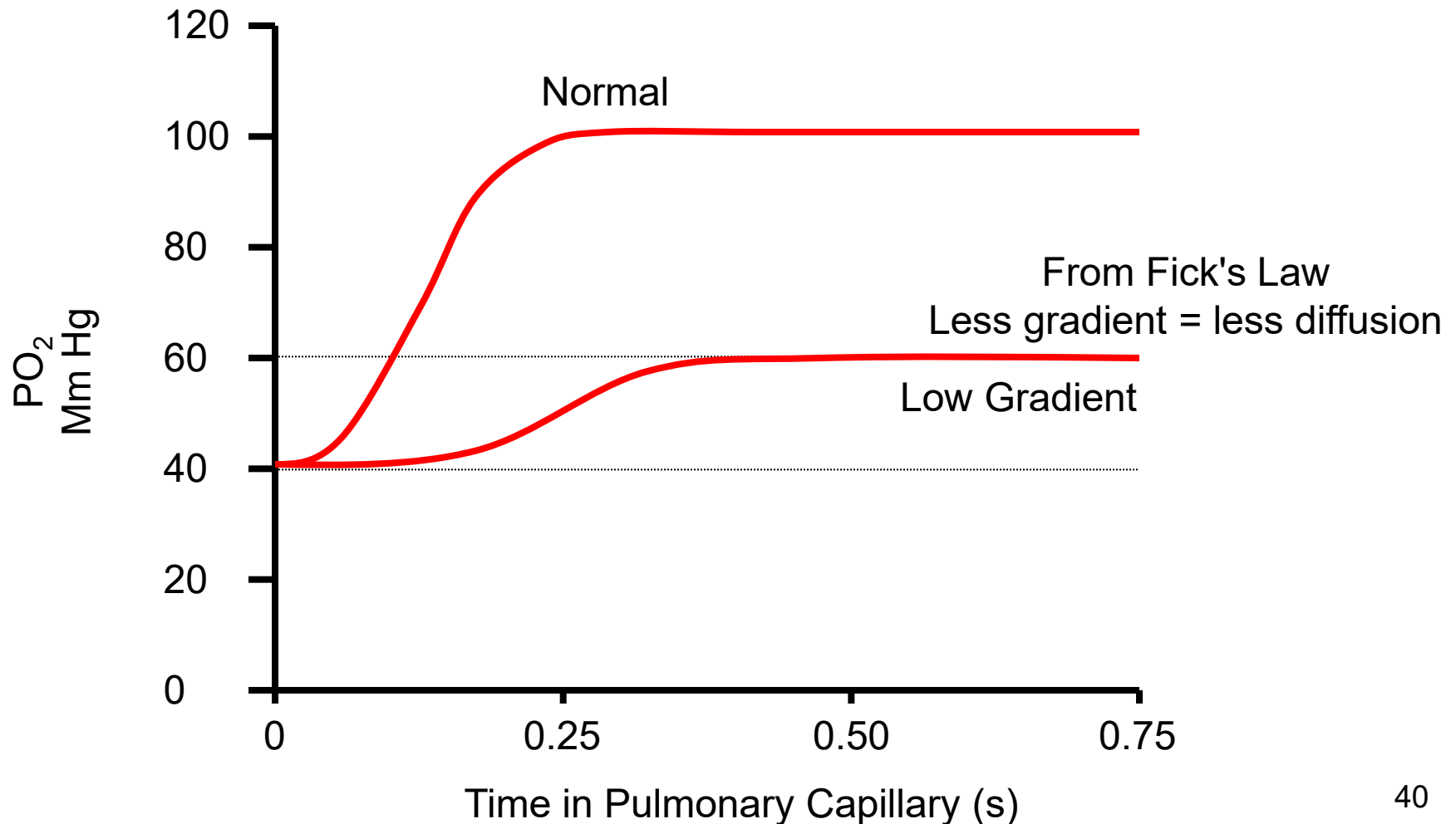
Increase barrier thickness -
reducing rate of diffusion



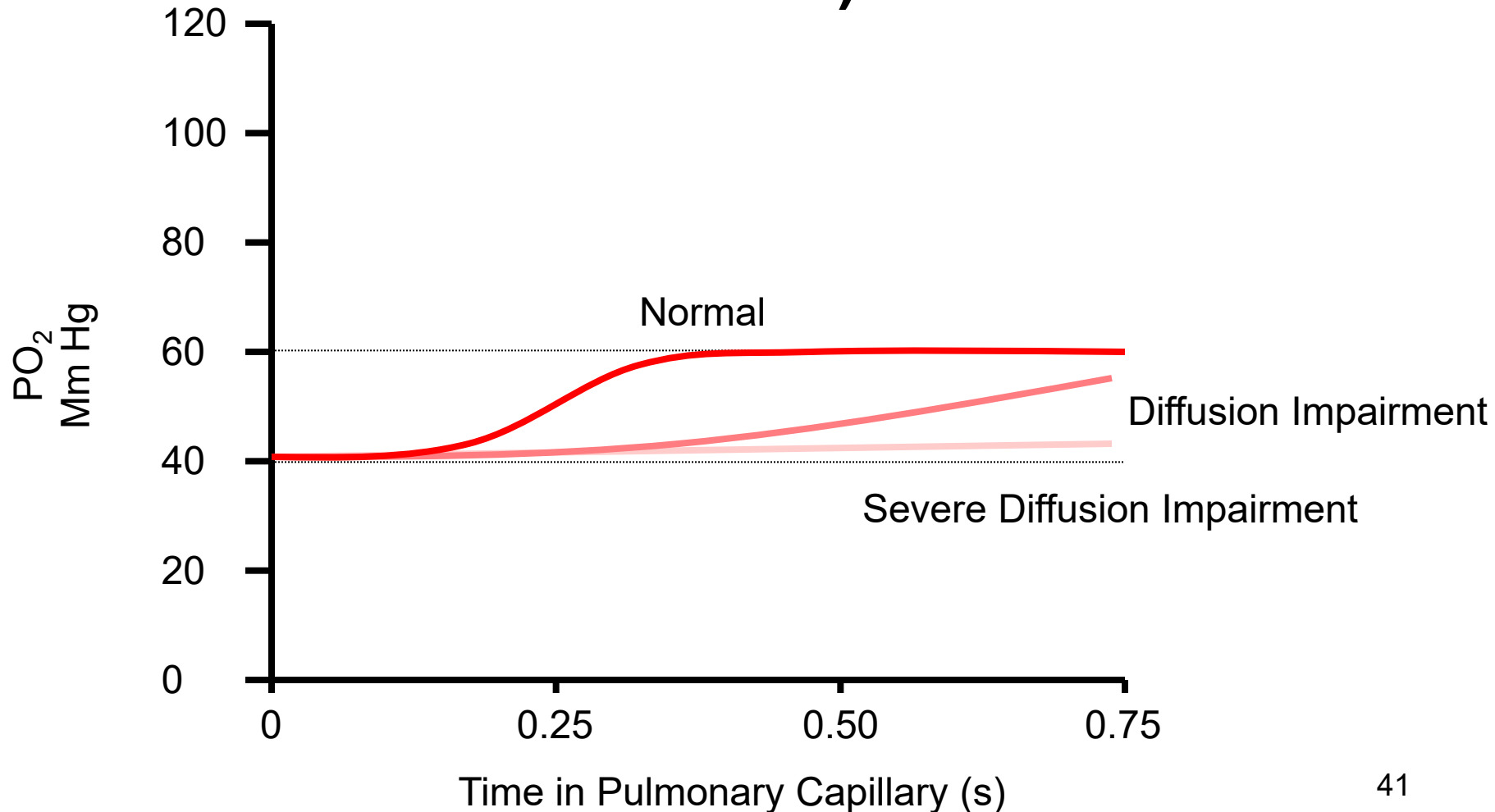
O₂ becomes Diffusion-limited in **lung diseases**:

- Pulmonary edema
- Pulmonary fibrosis

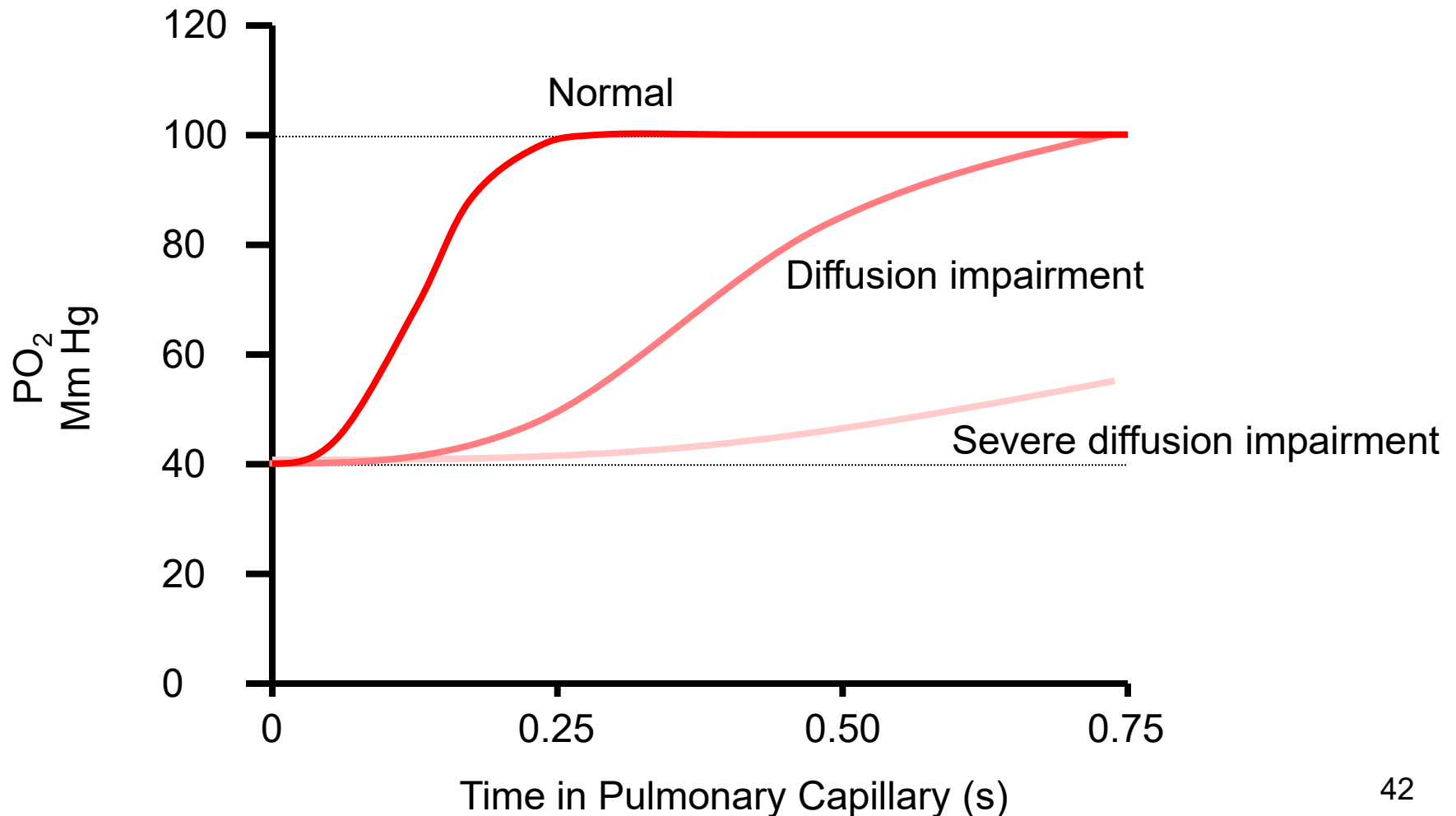
Oxygen Diffusion with Reduced O_2 Gradient (Example: Normal person at altitude)



Oxygen Diffusion with Reduced Oxygen Gradient (Patient having pulm edema at altitude)



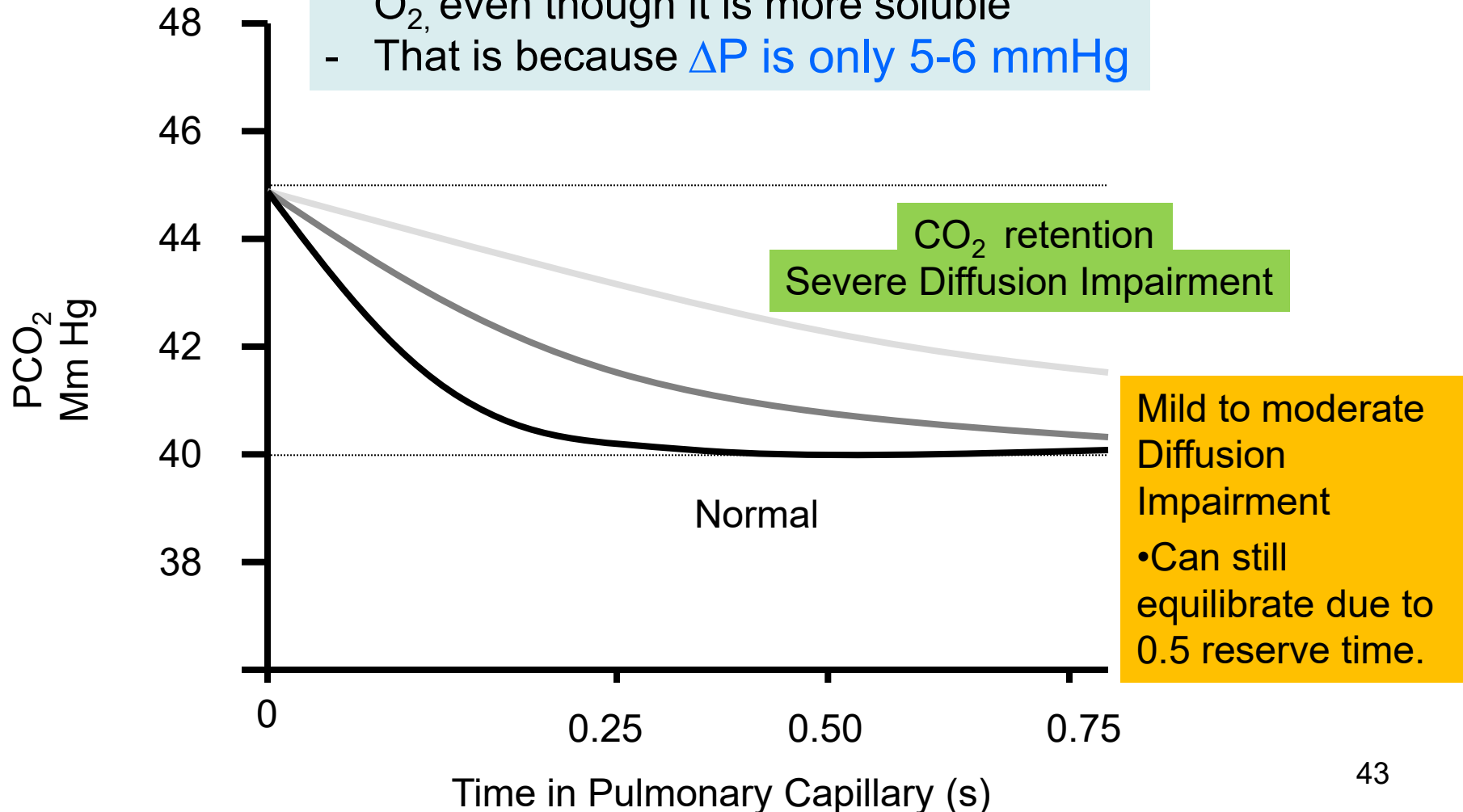
Oxygen Diffusion with Impaired diffusion (Example: patient with pulmonary fibrosis)



Carbon Dioxide Diffusion in lungs

CO₂ is perfusion-limited.

- Takes 0.25 secs to equilibrate, same as O₂, even though it is more soluble
- That is because ΔP is only 5-6 mmHg



Differences between Perfusion & Diffusion Limited gases

Perfusion limited

- There **is** equilibration of gas between alveolar & pulmonary capillary

Diffusion limited

- There is **no** equilibration of gas between alveolar & pulmonary capillary

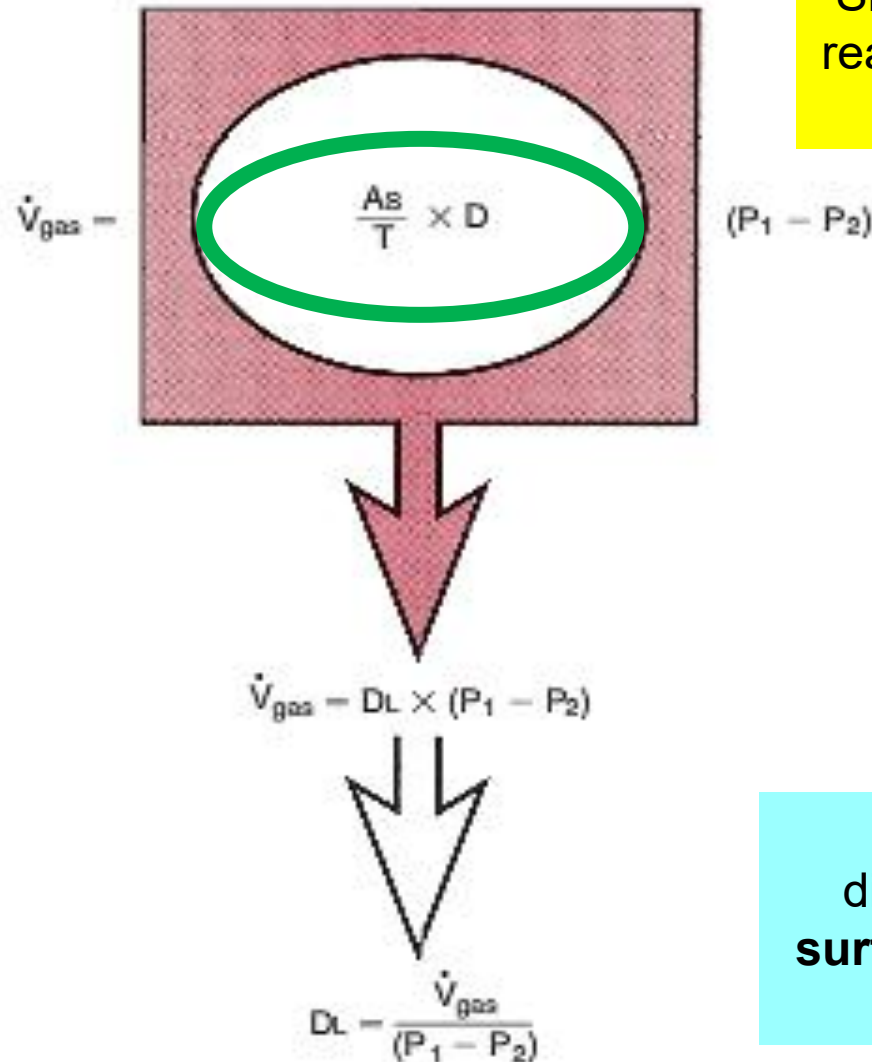
Diffusing Capacity (D_L)

- D_L is used to assess lung's ability to transfer gases.
- To measure D_L , a single breath test using carbon monoxide (CO).
- **Carbon monoxide** is chosen because it is
 - Diffusion-limited gas
 - None in pulmonary venous blood
 - Its high affinity to binding Hb makes $P_a\text{CO}$ near zero.

D_L is **decreased** in:

Patients with pulmonary fibrosis, pulmonary edema and loss of alveolar membrane (emphysema)

Measuring DL_{CO} - by Simplifying Fick's Law & using CO



Since D_L is hard to measure, you can use rearranged equation (below) for measuring D_L

$P_2 = P_{aCO}$ in pulmonary capillaries, which is nearly zero. Therefore,

$$D_L = V_{CO}/P_{ACO}$$

V_{CO} = CO uptake in mL/min

P_{ACO} = alveolar partial pressure of CO (P_1)

D_L : combined term for diffusion characteristics and includes:
surface area (A_s), membrane thickness (T) & the diffusion coefficient (D).

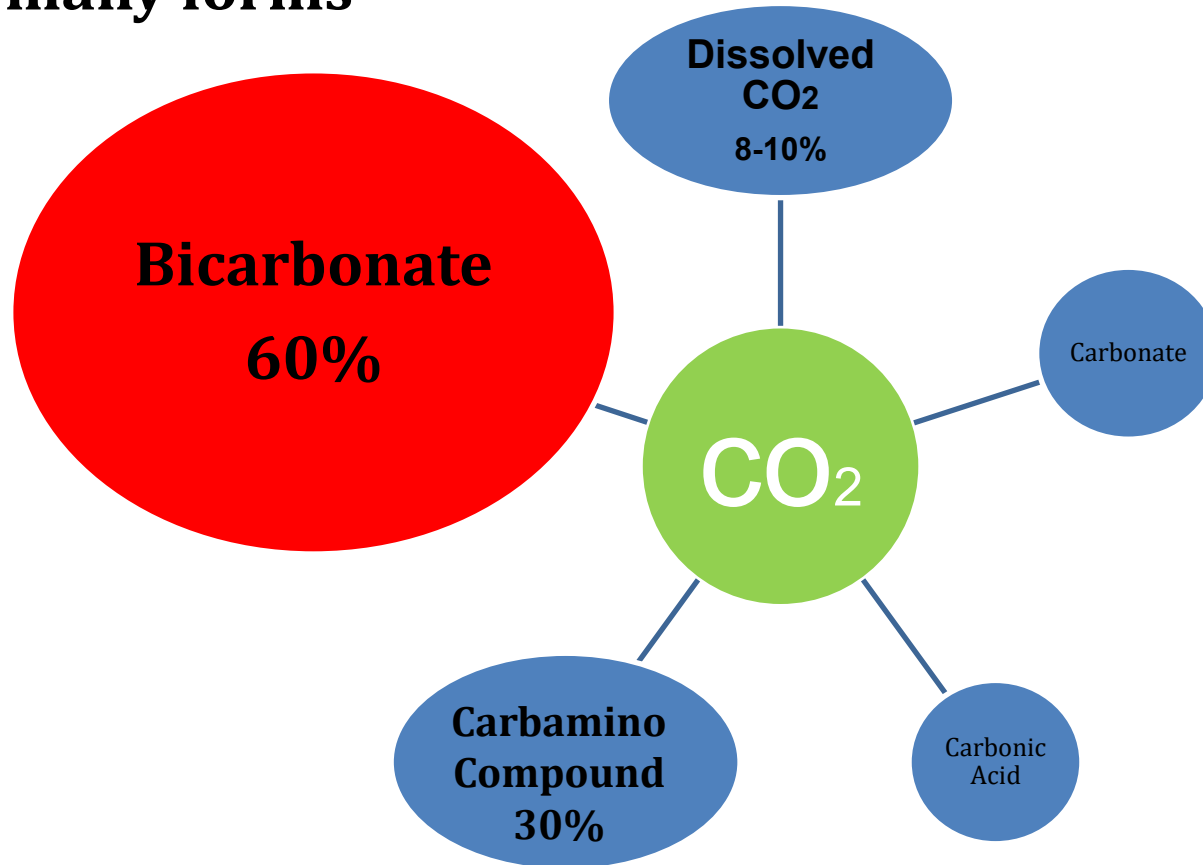
Single Breath Method of Assessing D_L

- Breath out to residual volume
- Take maximal inspiration with 0.3% CO/10% He mix of gas
- Hold breath for 10s
- Breath out – at end of breath measure CO and He
- Measure the difference between the concentration of He in inspired and expired air – this gives a dilution factor and can be used to calculate lung volume (since no He is taken up by lung)
- Measure the difference between the concentration of CO in inspired and expired air (corrected for the He dilution) & this will give you the amount of CO taken up by the lung in 10s
- Then use the equation $D_L = V_{CO}/P_A CO$
- Normal value of D_L for CO is 20-30 mL/min/mm Hg
- D_L is reduced ($\downarrow$) in pulmonary fibrosis, pulmonary edema (because of increased thickness) and in emphysema (because of loss of alveolar surface area).

Carbon Dioxide Transport

Carbon Dioxide Transport

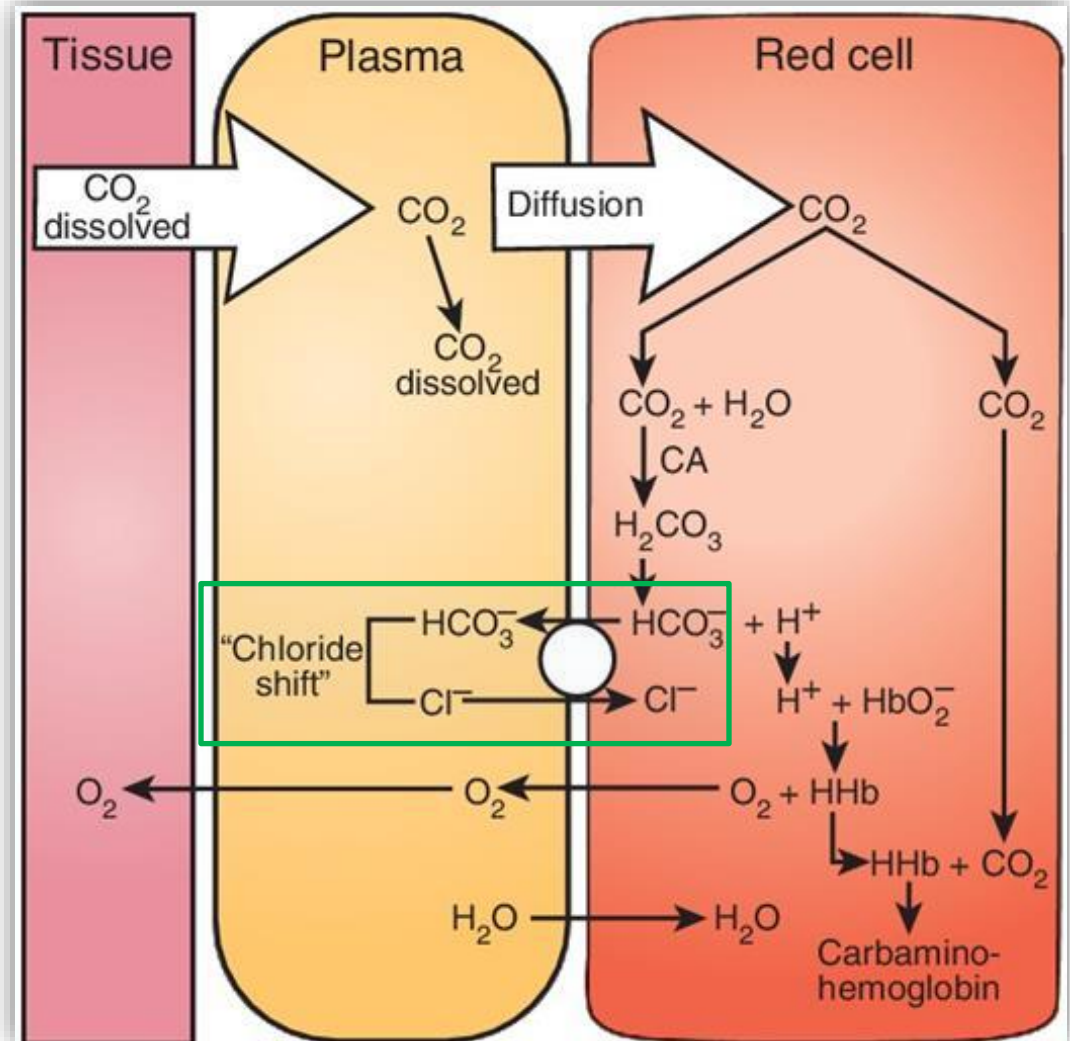
CO_2 is transported in many forms



Carbon Dioxide Transport: Biochemical reactions at tissue

Chloride shift: exchange of chloride for bicarbonate in RBCs for the maintenance of electrical neutrality

Reverse reactions occur at lungs to eliminate CO_2 to the atmosphere



Carbon Dioxide Transport

Carbonic Anhydrase



- The combination of CO_2 and H_2O occurs slowly in the absence of **carbonic anhydrase (CA)**
- This enzyme accelerates this reaction by about 1,000 times
- This enzyme is found in high concentration in red blood cells.

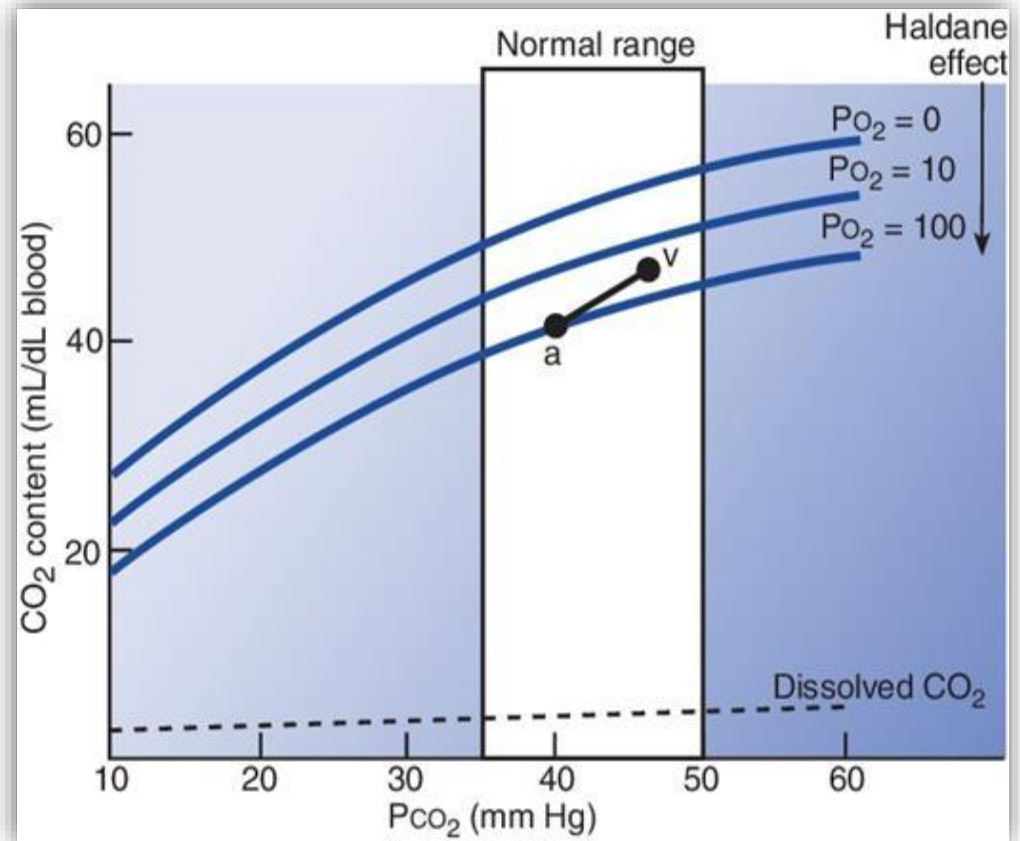
Carbon Dioxide Transport

CO₂ Content

- Relatively linear relationship with PaCO₂
- CO₂ is more soluble in plasma than O₂
- Large amounts of CO₂ can be loaded and unloaded from the blood with a small change in PCO₂
- Increasing the PO₂ causes the blood to unload CO₂ (where do we see this?)
- Blood returning to lungs offloads CO₂ as it picks up O₂

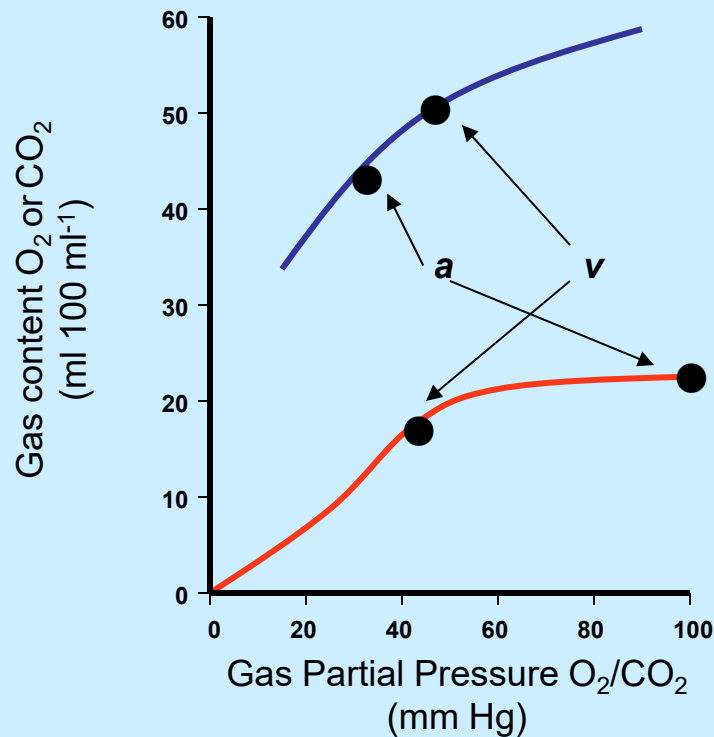
Carbon Dioxide Transport - Haldane Effect

- The effect of the PO_2 on the CO_2 equilibrium curve is known as the **Haldane effect**
- Higher PO_2 will shift the curve downward and to the right.
- This allows the blood to load more CO_2 in the tissues and unload more CO_2 in the lungs.



The carbon dioxide equilibrium curve is relatively linear. **An increase in PO_2 tension causes a rightward and downward shift of the curve.** a, arterial CO_2 content; v, CO_2 content in mixed venous blood.

O₂ and CO₂ curves: Comparison



**a, arterial blood;
v, mixed venous
blood.**

1. CO₂ content much higher than O₂ content. 1 liter of blood can hold much more CO₂ than O₂.

2. For CO₂, small change in PCO₂ can load or unload greater amount of CO₂ as the gradient of the curve is steeper & linear.

Carbon Dioxide Transport

Carbonic Anhydrase



- Hydrogen ion concentration, bicarbonate concentration and PCO₂ are all inter-related
- Carbonic acid (H₂CO₃), although relatively weak can alter pH (**normal 7.35 - 7.45**)

Henderson Hasselbach Equation

$$pH = pK + \log \frac{[\text{HCO}_3^-]}{.03 \cdot \text{PCO}_2}$$

Carbon Dioxide Transport & pH

- The reference range of the pH of arterial blood is 7.35-7.45.
- **Respiratory Acidosis** - condition of lowered pH (acidosis) due to decreased alveolar ventilation (hypoventilation and $\uparrow\text{PaCO}_2$) as seen in an acute exacerbation of asthma.
- **Respiratory Alkalosis** - condition of increased pH (alkalosis) due to increased alveolar ventilation as seen at high altitude.
(Remember hyperventilation: $\downarrow\text{PaCO}_2$).

Any Questions ?

Please reach me: Either walk into my office or email.

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